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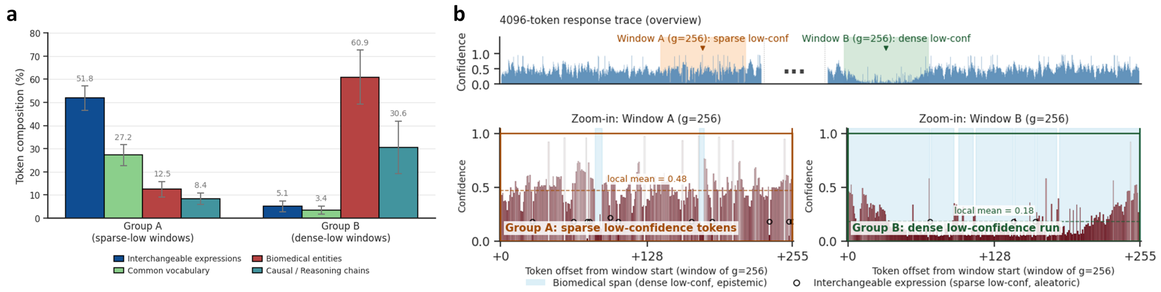
Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

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Biomedical uncertainty has local structure

Dense low-confidence runs are enriched for biomedical entities and causal chains. Absolute-confidence weighting can suppress these learning signals.



PMC: biomedical entities account for 60.9% of low-confidence tokens in dense windows, versus 12.5% in sparse windows. Expert CRAFT annotations corroborate concept enrichment (63.8% versus 11.2%).

Balanced Fine-Tuning at two scales

Token weight: relative to the local context

$$\rho_{\beta,t} = \mathrm{clip}(c_{\beta,t} / (C_{\beta,t}^{\mathrm{loc}} + \varepsilon), 0, 1)$$

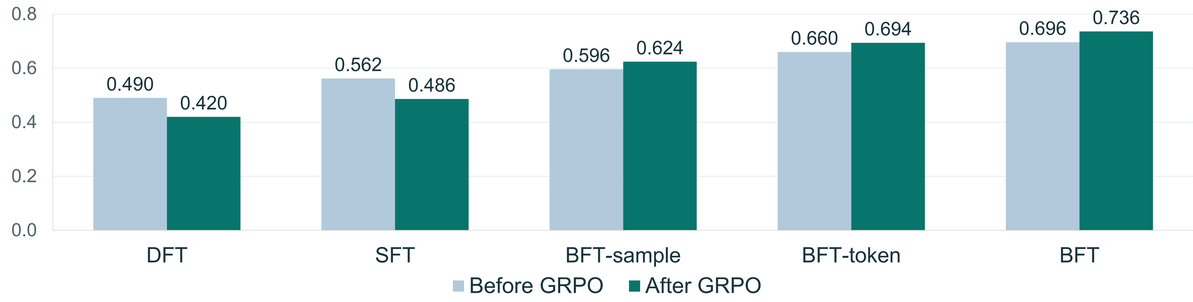
Sample weight: emphasize the weakest region

$$s_\beta = \mathrm{sg}(1 - \min_t C_{\beta,t}^{\mathrm{loc}})$$

The loss averages $s_\beta \times \mathrm{sg}(\rho_{\beta,t}) \times$ token cross-entropy over valid response tokens, then across samples. Both weights use stop-gradient. The centered window g = 256 is the only additional tunable train-time hyperparameter.

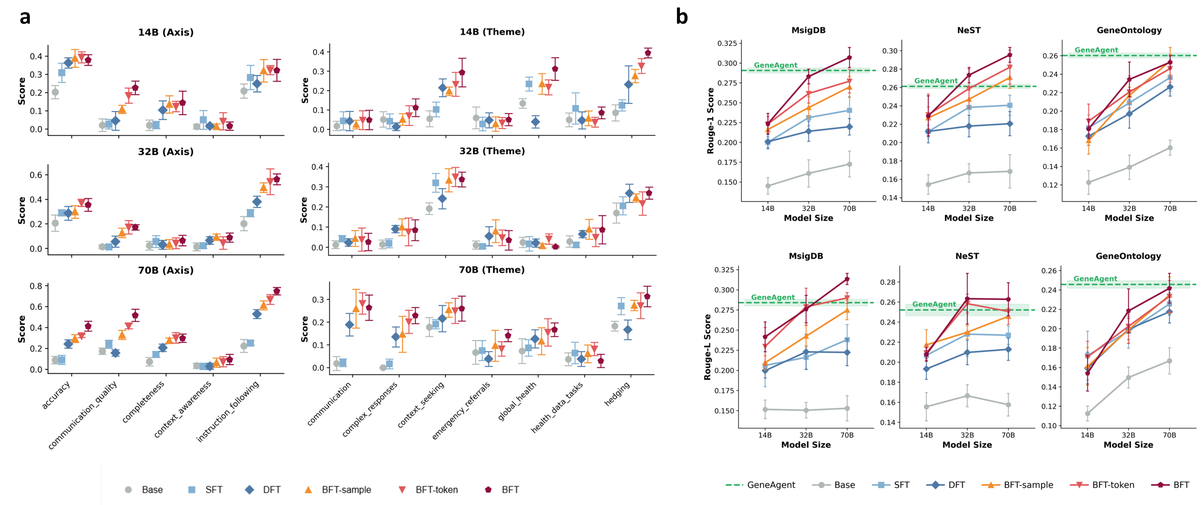
BFT supports subsequent sparse-reward RL

VCWorld chemical perturbation accuracy, 70B backbones.
Shared GRPO setup, five held-out cancer cell lines.



BFT: 0.696 to 0.736 (+4.0 pp). SFT: 0.562 to 0.486.
BFT is numerically above Gemini-2.5-Flash before GRPO, but the difference is not significant (p = 0.088).

Medical competence and biological reasoning



HealthBench Hard and GeneAgent: more consistent improvements under a shared supervised training recipe across 14B, 32B and 70B.

Generated profiles yield useful representations

BFT-generated profiles, encoded with text-embedding-ada-002, give the best average gene-level performance and improve evaluated cell clustering. With the STATE decoder retained, perturbation performance is comparable to standard inputs across four datasets.

Local relative confidence preserves difficult tokens in difficult contexts.

Paper: arxiv.org/abs/2511.21075
Code: github.com/TencentAILabHealthcare/BFT